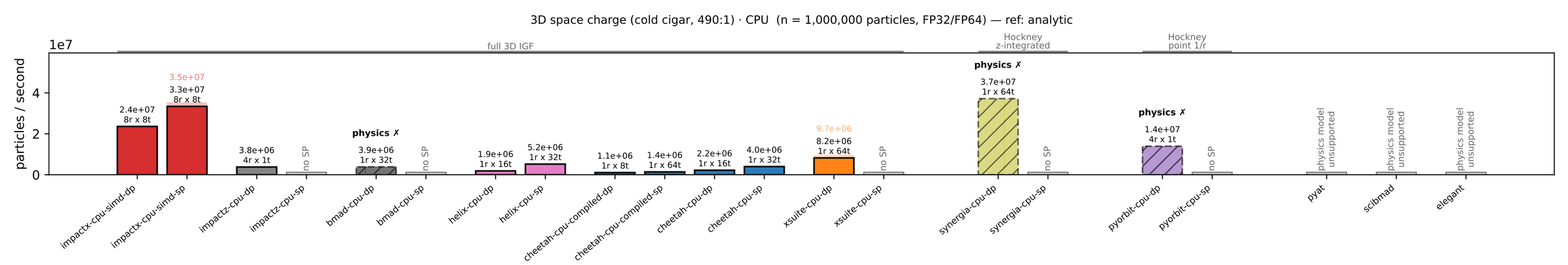

'physics ✗' = converged but out of tolerance

lighter bar behind + lighter value above = fast-math (relaxed FP), shown when >=5% faster than the IEEE bar

kernel over a mesh cell: IGF = integrated in 3D; z-integrated = along z only, point-sampled in x,y; point 1/r = sampled at cell centres

versions: impactx 26.08 · impactz V2.7.7 · bmad 20260904-1 · helix v1.10.1 · cheetah 0.8.2 · xsuite 0.53.4 · synergia v2024.04-209-g795b99f39 · pyorbit 96c8def · pyat 0.8.0 · scibmad 0.4.1 · elegant elegant-2026.4.0